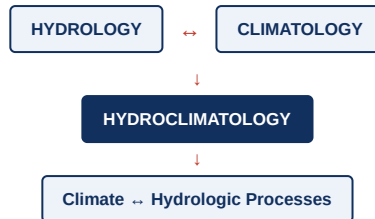


Lecture 3 — Role of Statistical & Machine Learning Methods in Interdisciplinary Studies

Compact revision sheet — concepts strictly as covered in lecture

1 Hydroclimatology — The Interdisciplinary Example



HYDROLOGY

Science of occurrence & movement of water (solid/liquid/vapor) across **surface, subsurface & atmosphere**. Based on the hydrologic (water) cycle. Manages water use, water control & water pollution.

CLIMATOLOGY

Study of exchange of **mass, momentum & energy** between land/ocean surface and atmosphere. Vertical & horizontal fluxes drive surface–atmosphere interaction and control atmospheric circulation at different scales.

HYDROCLIMATOLOGY

Interdisciplinary study of the interaction between hydrology & climatology — identifies influence of the climatic system on hydrologic processes within the hydrologic cycle. Atmospheric hydrologic cycle is coupled with climatic phenomena; variability in one affects the other via feedback.

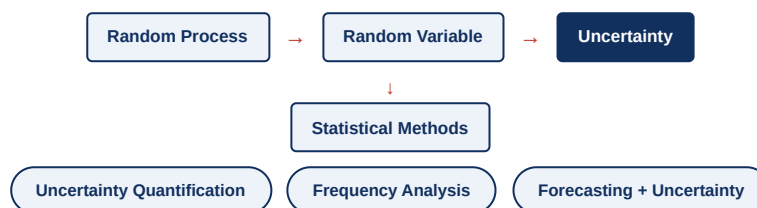
EXAMPLE Atmospheric circulation influences **rainfall**, which affects **soil moisture**, which affects **streamflow** — global/local circulation can significantly shift these hydrologic variables.



2 Why Statistics & ML Matter

- Widely used to gain valuable **insights across various fields**.
- Provide methods to **combine heterogeneous datasets** and **quantify uncertainty**.
- In scientific/engineering fields, core physical mechanisms are analyzed depending on required **detail & scale**.
- Data from other fields can be used to reveal insights into the underlying physical process.

3 Random Process & Uncertainty

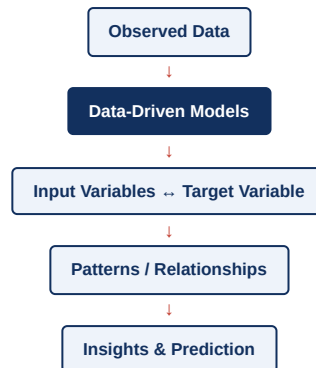


EXAMPLE Annual maximum **rainfall** / annual maximum **flow** vary year to year — uncertainty in these hydrologic variables is treated via statistical methods.

Lecture 3 — Combined Role, Curse of Dimensionality & Revision

Continued from Page 1

4 Combined Role of Statistical & ML Methods



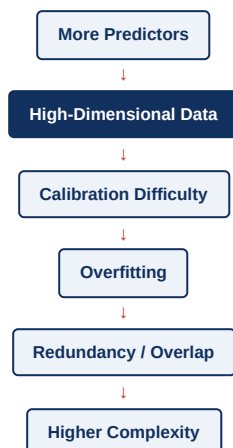
- **Data-driven models** infer patterns from observed data instead of explicitly encoding physical laws; they explore relationships between input variables & the target variable.
- Reliable prediction needs the **best set of predictors** — predictor selection depends on physical laws / domain understanding.
- **Domain knowledge** + combined statistical/ML approaches are needed to capture complex underlying associations.

EXAMPLE With rainfall & other observed variables as inputs and streamflow as target, a data-driven model learns the input–target relationship from data rather than encoding every physical law.

Key Takeaway

Data-driven methods discover patterns from data, while domain knowledge helps identify meaningful predictors and improve reliable prediction.

5 Curse of Dimensionality



- **Curse of dimensionality**: phenomena arising when analyzing/organizing data in high-dimensional spaces, not seen in low-dimensional settings.
- Large predictor set → hard to calibrate model, may cause **overfitting**.
- Highly correlated predictors convey **overlapping information** — a second input adds little new info but raises complexity & redundancy.

6 Dimensionality Reduction



FEATURE ENGINEERING

Construct / use useful representations of predictors.

FEATURE SELECTION

Select useful predictors, reduce unnecessary inputs.

PRINCIPAL COMPONENT ANALYSIS (PCA)

Named in the lecture as a dimensionality-reduction method (combination of statistical & ML methods).

LECTURE 3 — QUICK REVISION

1. **Hydroclimatology** = interaction between hydrology and climatology.
2. Hydrologic variables include rainfall, soil moisture, and streamflow.
3. Statistics/ML help combine heterogeneous data and quantify uncertainty.
4. Statistical methods address uncertainty, frequency analysis, and forecasting with uncertainty.
5. Data-driven models learn relationships between inputs and targets from observed data.
6. Reliable prediction requires appropriate predictors and domain/physical knowledge.
7. High-dimensional predictor sets can cause calibration difficulty, overfitting, redundancy, and complexity.
8. Feature Engineering, Feature Selection, and PCA can help with dimensionality reduction.